Technical System Specification & Architecture Manual

Hybrid multimodal fall detection vision based and sensor based & emergency dispatch system (HMFDA)

Technical specification | Architecture | APIs | Verification

SOURCE DOCUMENTATION.md

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1. Executive Summary & Problem Statement

Falls represent the leading cause of accidental injury and trauma in elderly populations globally. Conventional fall detection systems typically suffer from two severe limitations:

1. Single-modality blind spots: Wearable-only systems are frequently forgotten or left on charging docks; camera-only systems cannot protect the individual outside the home or in unmonitored zones.
2. False alarms and high latency: Sudden sitting, lying down, or dropping a phone causes false alarms that exhaust emergency services and caretakers.

HMFDA addresses these challenges via a hybrid multimodal edge-cloud architecture:

1. Indoor Spaces: Non-intrusive ambient vision powered by YOLO11n-Pose (extracting 17 human body keypoints) and temporal Gated Recurrent Units (GRU) with temporal confirmation.
1. Outdoor Spaces: Smartphone-based 9-channel inertial tracking (accelerometer, gravity, gyroscope) evaluated simultaneously by a Bidirectional Keras LSTM and on-device kinematic drop-shock physics heuristics.
1. Cloud Brokerage: A centralized Alert Manager microservice providing idempotency, source-based cooldown, and an active alarm hold window.
1. Emergency Dispatch: Automatic audible siren with an 8-second false alarm cancellation window, hands-free background SMS with live Google Maps GPS coordinate link, and direct cellular phone calls.

2. Mathematical Formulations & Pipeline Mechanics

2.1 Indoor Vision Pipeline (YOLO11n-Pose + GRU)

A. Keypoint Extraction

Given an input video frame $I_t \in \mathbb{R}^{H \times W \times 3}$ at time t, the YOLO11n-Pose backbone extracts K = 17 keypoints corresponding to the standard COCO human skeleton:

$$\mathcal{P}_t = \{(x_{i,t}, y_{i,t}, c_{i,t})\}_{i=1}^{17}$$

where (x_i, y_i) are image pixel coordinates and $c_i \in [0, 1]$ represents landmark confidence.

B. Spatial Coordinate Normalization

To make the pose invariant to the subject's distance from the camera and physical height:

1. Calculate the mid-hip anatomical anchor:

$$\bar{x}_{\text{hip}} = \frac{x_{11,t} + x_{12,t}}{2}, \quad \bar{y}_{\text{hip}} = \frac{y_{11,t} + y_{12,t}}{2}$$

1. Compute the bounding box height $H_{bbox} = y_{max} - y_{min}$.
2. Shift and scale all keypoints:

$$\hat{x}_{i,t} = \frac{x_{i,t} - \bar{x}_{\text{hip}}}{H_{\text{bbox}}}, \quad \hat{y}_{i,t} = \frac{y_{i,t} - \bar{y}_{\text{hip}}}{H_{\text{bbox}}}$$

```
\hat{x}_{i,t} = \frac{x_{i,t} - \bar{x}_{\text{hip}}}{H_{\text{bbox}}}, \quad \hat{y}_{i,t} = \frac{y_{i,t} - \bar{y}_{\text{hip}}}{H_{\text{bbox}}}
```

The normalized frame vector is v_t = [x̂_1,t, ŷ_1,t, …, x̂_17,t, ŷ_17,t]^T □ □^34.

C. Temporal Sequence Modeling via GRU

A sliding window of T_vision = 20 consecutive frames forms the temporal sequence:

$$\mathbf{X}_{\text{vision}} = [\mathbf{v}_{t-19}, \mathbf{v}_{t-18}, \dots, \mathbf{v}_t] \in \mathbb{R}^{20 \times 34}$$

```
\mathbf{X}_{\text{vision}} = [\mathbf{v}_{t-19}, \mathbf{v}_{t-18}, \dots, \mathbf{v}_t] \in \mathbb{R}^{20 \times 34}
```

For each step τ □ 1, …, 20, the GRU cell updates its hidden state h_τ:

$$z_\tau = \sigma(\mathbf{W}_z \mathbf{v}_\tau + \mathbf{U}_z h_{\tau-1} + b_z)$$
$$r_\tau = \sigma(\mathbf{W}_r \mathbf{v}_\tau + \mathbf{U}_r h_{\tau-1} + b_r)$$
$$\tilde{h}_\tau = \tanh(\mathbf{W}_h \mathbf{v}_\tau + \mathbf{U}_h (r_\tau \odot h_{\tau-1}) + b_h)$$
$$h_\tau = (1 - z_\tau) \odot h_{\tau-1} + z_\tau \odot \tilde{h}_\tau$$

```
\begin{aligned} z_\tau &= \sigma\left(\mathbf{W}_z \mathbf{v}_\tau + \mathbf{U}_z h_{\tau-1} + b_z\right) \quad &&\text{(Update gate)} \\ r_\tau &= \sigma\left(\mathbf{W}_r \mathbf{v}_\tau + \mathbf{U}_r h_{\tau-1} + b_r\right) \quad &&\text{(Reset gate)} \\ \tilde{h}_\tau &= \tanh\left(\mathbf{W}_h \mathbf{v}_\tau + \mathbf{U}_h (r_\tau \odot h_{\tau-1}) + b_h\right) \quad &&\text{(Candidate state)} \\ h_\tau &= (1 - z_\tau) \odot h_{\tau-1} + z_\tau \odot \tilde{h}_\tau \quad &&\text{(Hidden state)} \end{aligned}
```

The final hidden state h_20 is mapped to the fall probability:

$$p_{\text{vision}} = \sigma(\mathbf{W}_o h_{20} + b_o)$$

```
p_{\text{vision}} = \sigma\left(\mathbf{W}_o h_{20} + b_o\right)
```

D. Multi-Frame Confirmation Rule

To suppress sudden motions (e.g., rapidly sitting on a chair), an alert is generated if and only if:

$$\sum_{k=0}^{M-1} \mathbb{I}(p_{\text{vision},\,t-k} \geq \tau_{\text{vision}}) = M$$

```
\sum_{k=0}^{M-1} \mathbb{I}\left(p_{\text{vision},\,t-k} \ge \tau_{\text{vision}}\right) = M
```

where threshold τ_vision = 0.35 and confirmation length M = 3 consecutive frames.

2.2 Outdoor Smartphone Inertial Pipeline (9-Channel LSTM + Kinematics)

A. 9-Channel Sensor Representation

The smartphone streams 3 hardware sensors sampled at f_s = 50 Hz:

$$\mathbf{S}_t = \begin{matrix} \mathbf{a}_t \\ \mathbf{g}_t \\ \boldsymbol{\omega}_t \end{matrix} = \begin{matrix} [a_{x,t}, a_{y,t}, a_{z,t}]' \\ [g_{x,t}, g_{y,t}, g_{z,t}]' \\ [\omega_{x,t}, \omega_{y,t}, \omega_{z,t}]' \end{matrix}$$

(Linear Acceleration) | (Gravitational Vector) | (Angular Velocity)

$$\in \mathbb{R}^9$$

```
\mathbf{S}_t = \begin{bmatrix} \mathbf{a}_t \\ \mathbf{g}_t \\ \boldsymbol{\omega}_t \end{bmatrix} = \begin{bmatrix} [a_{x,t}, a_{y,t}, a_{z,t}]^T & \text{(Linear Acceleration)} \\ [g_{x,t}, g_{y,t}, g_{z,t}]^T & \text{(Gravitational Vector)} \\ [\omega_{x,t}, \omega_{y,t}, \omega_{z,t}]^T & \text{(Angular Velocity)} \end{bmatrix} \in \mathbb{R}^9
```

B. Statistical Standardization

Continuous channels are normalized using the pre-computed SisFall training dataset statistics:

$$z_{t,k} = \frac{s_{t,k} - \mu_k}{\sigma_k}, \quad k \in \{1, \dots, 9\}$$

```
z_{t, k} = \frac{s_{t, k} - \mu_k}{\sigma_k}, \quad k \in \{1, \dots, 9\}
```

where μ_k and σ_k are the mean and standard deviation of channel k.

C. Temporal Sliding Window & LSTM Inference

A FIFO buffer stores N = 200 samples (equivalent to a 4.0-second temporal window at 50Hz):

$$\mathbf{X}_{\text{sensor}} = [\mathbf{z}_{t-199}, \dots, \mathbf{z}_t] \in \mathbb{R}^{200 \times 9}$$

```
\mathbf{X}_{\text{sensor}} = [\mathbf{z}_{t-199}, \dots, \mathbf{z}_t] \in \mathbb{R}^{200 \times 9}
```

The Bidirectional LSTM processes the forward and backward representations:

$$\vec{h}_\tau = \text{LSTM}_{\text{fwd}}(\mathbf{z}_\tau, \vec{h}_{\tau-1}), \quad \overleftarrow{h}_\tau = \text{LSTM}_{\text{bwd}}(\mathbf{z}_\tau, \overleftarrow{h}_{\tau+1})$$

```
\vec{h}_\tau = \text{LSTM}_{\text{fwd}}(\mathbf{z}_\tau, \vec{h}_{\tau-1}), \quad \overleftarrow{h}_\tau = \text{LSTM}_{\text{bwd}}(\mathbf{z}_\tau, \overleftarrow{h}_{\tau+1})
```

$$h_{\text{combined}} = [\vec{h}_{200} \,\|\, \overleftarrow{h}_1]$$

```
h_{\text{combined}} = [\vec{h}_{200} \,\|\, \overleftarrow{h}_1]
```

$$p_{\text{lstm}} = \sigma(\mathbf{W}_{\text{dense}} h_{\text{combined}} + b_{\text{dense}})$$

```
p_{\text{lstm}} = \sigma\left(\mathbf{W}_{\text{dense}} h_{\text{combined}} + b_{\text{dense}}\right)
```

D. Physical Kinematics Impact Engine

In parallel with the neural network, physical acceleration magnitude is tracked across the most recent 60 samples (sim 1.2s window):

$$\mathcal{A}_t = \sqrt{a_{x,t}^2 + a_{y,t}^2 + a_{z,t}^2}$$

```
\mathcal{A}_t = \sqrt{a_{x,t}^2 + a_{y,t}^2 + a_{z,t}^2}
```

1. Minimum Acceleration (Free-fall dip): a_min = min_τ ∈ [t-60, t] A_τ
1. Maximum Acceleration (Impact spike): a_max = max_τ ∈ [t-60, t] A_τ

Kinematic Trigger Rule:

$$\text{KinematicFall} = (a_{\max} \geq 14.0\ \text{m/s}^2 \land a_{\min} \leq 6.8\ \text{m/s}^2) \lor (a_{\max} \geq 19.0\ \text{m/s}^2)$$

```
\text{KinematicFall} = \Big(a_{\max} \ge 14.0\text{ m/s}^2 \land a_{\min} \le 6.8\text{ m/s}^2\Big) \lor \Big(a_{\max} \ge 19.0\text{ m/s}^2\Big)
```

$$\text{FinalOutdoorFall} = \text{KinematicFall} \lor (p_{\text{lstm}} \geq 0.40)$$

```
\text{FinalOutdoorFall} = \text{KinematicFall} \lor \Big(p_{\text{lstm}} \ge 0.40\Big)
```

3. Microservice Architectural Specification & API Contracts

3.1 Network Topology & Port Bindings

Service	Container Name	Bind Address	Internal / External Port	Protocol
Gateway	`hmfda-gateway`	`0.0.0.0`	`80 / 80`	HTTP / WebSocket
Alert Manager	`hmfda-alert-manager`	`0.0.0.0`	`8002 / 8002`	HTTP / JSON
Sensor Service	`hmfda-sensor-service`	`0.0.0.0`	`8000 / 8000`	HTTP / JSON
Dashboard	`hmfda-dashboard`	`0.0.0.0`	`8501 / 8501`	HTTP / WebSocket

3.2 Central Alert Manager API Contracts (`Port 8002`)

The Alert Manager coordinates between detectors and mobile devices.

1. Ingest Confirmed Fall Event

1. Endpoint: `POST /event`
1. Headers: `Content-Type: application/json`
1. Request Payload:

```
{
  "source": "indoor", // or "outdoor"
  "event_id": "c1f7b830-672e-4b68-968b-59b5be2d8a43",
  "timestamp": "2026-10-23T04:15:30.000Z",
  "prediction": "FALL",
  "probability": 0.965
}
```

1. Success Response (`200 OK`):

```
{
  "accepted": true,
  "event_id": "c1f7b830-672e-4b68-968b-59b5be2d8a43",
  "hold_seconds": 20.0
}
```

1. Deduplication Response (`200 OK`):

```
{
```

```
  "accepted": false,
  "reason": "duplicate event_id or inside 15s cooldown",
  "duplicate": true
}
```

2. Query System Status (Mobile Polling & Dashboard)

- Endpoint: `GET /status`
- Response Payload:

```
{
  "probability": 0.965,
  "fall": true,
  "samples": 200,
  "connected": true,
  "source": "outdoor",
  "last_event": {
    "source": "outdoor",
    "event_id": "c1f7b830-672e-4b68-968b-59b5be2d8a43",
    "timestamp": "2026-10-23T04:15:30.000Z",
    "prediction": "FALL",
    "probability": 0.965
  },
  "pending_events": 1
}
```

3. Smartphone Sensor Batch Relay

- Endpoint: `POST /`
- Relay Target: Forwards raw payload to `OUTDOOR_DETECTOR_URL` (`http://sensor-service:8000/`).
- Response: Returns the live evaluated probability and rolling buffer sample count.

3.3 Outdoor Sensor Microservice API (`Port 8000`)

- Endpoint: `POST /`
- Receives 9-channel sensor batches (Sensor Logger format).
- Appends samples to rolling 200-sample queue.
- Evaluates LSTM model + kinematic thresholds.
- Automatically triggers `POST http://alert-manager:8002/event` if a fall is detected.
- Endpoint: `GET /status`
- Returns: `{"probability": 0.002, "threshold": 0.4, "fall": false, "samples": 200, "connected": true}`
- When a fall occurs, holds `"fall": true` and `"probability": 0.965` for 20 seconds.

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4. Mobile Application Architecture (`FallAlert.apk`)

```
com.fallalert.app
├── MainActivity.kt              # Entrypoint & Runtime Permission Dispatcher
├── data/
│   └── ContactRepository.kt     # Encrypted SharedPreferences (Emergency Contact, Server URL)
├── sensor/
│   ├── SensorCollector.kt       # 50Hz SensorEventListener (Acc, Grav, Gyro) + Kinematics
Engine
│   ├── DetectionNetworkClient.kt # Non-blocking HttpURLConnection client (POST /, GET /status)
│   └── LocationHelper.kt        # FusedLocationProviderClient (High-accuracy GPS updates)
├── alert/
│   ├── AlertController.kt       # MediaPlayer siren loop + Vibrator motor pulsing
│   ├── EmergencyNotifier.kt     # SmsManager dispatcher + Intent.ACTION_CALL / ACTION_DIAL
│   └── FallAlertState.kt        # StateFlow data models (UI state, emergency events)
└── ui/
    ├── FallAlertViewModel.kt    # State orchestration, countdown coroutines, dispatch state
machine
    ├── screens/HomeScreen.kt    # Jetpack Compose dark-mode UI with live telemetry
    └── theme/                   # Material3 Design Tokens (AlertRed, SurfaceDark, Typography)
```

Emergency Dispatch State Machine

```
stateDiagram-v2
    [*] --> Monitoring : User taps "Start Detection"
    Monitoring --> FallDetected : Kinematic shock OR Cloud fall=true

    state FallDetected {
        [*] --> CountdownActive : Start Siren & Vibration
        CountdownActive --> Dispatched : 20 seconds expire without cancellation
        CountdownActive --> Cancelled : User taps "Stop Alarm"
    }

    Cancelled --> Monitoring : System re-armed

    state Dispatched {
        [*] --> SendSMS : Dispatches SMS via device SIM
        SendSMS --> DialCall : Places direct cellular call
        DialCall --> LocationLoop : GPS link every 30s
    }

    Dispatched --> Monitoring : User silences alarm
```

5. Production AWS Cloud Infrastructure

Directory Structure on AWS EC2 (`/home/ubuntu/HMFDA`)

```
~/HMFDA/
├── docker-compose.yml              # Production container orchestration
├── .env                            # Secret credentials (Twilio, Ports, URLs)
├── deploy-aws.sh                   # Automated provisioning script
├── docker/
│   ├── Dockerfile.backend          # Alert Manager & Sensor Service image
│   ├── Dockerfile.dashboard        # Streamlit 3D Digital Twin image
│   ├── nginx.conf                  # Nginx reverse proxy configuration
│   └── web/
│       ├── index.html              # Cloud Gateway portal landing page
│       └── download/
│           └── fallalert.apk       # Downloadable Android companion package
├── Hybrid-Multimodel-Fall-Detection-System/
│   ├── sensor_service.py           # Multi-threaded Keras LSTM service
│   ├── app/
│   │   ├── app.py                  # Streamlit application
│   │   ├── alert_system.py         # Caretaker and Twilio notification logic
│   │   └── three_d_components.py   # 3D Room digital twin generator
│   ├── models/
│   │   ├── outdoor/best_outdoor_lstm.keras
│   │   └── vision/yolo11n-pose.pt
│   └── src/
│       └── alerting/alert_manager.py
```

6. End-to-End Verification & Verification Matrix

Test Scenario	Input Action	Expected System Response	Verification Status
Indoor Fall	Actor drops onto floor in front of webcam	`GRU confirms 3 frames ≥ 0.35; sends POST /event; Dashboard turns red`	PASS
Indoor Sitting	Actor sits down quickly in chair	Pose aspect ratio shifts but confirmation fails; status stays safe	PASS
Outdoor Phone Drop	Phone dropped onto bed / sofa	Accelerometer dips <6.8 m/s^2 and spikes >14.0 m/s^2; siren starts	PASS
False Alarm Cancel	User taps "Stop Alarm" within 20s	Siren stops immediately; no SMS or call dispatched; system re-arms	PASS
Real Fall Emergency	Phone dropped; 20s timer allowed to expire	Phone sends SMS with Google Maps link + dials phone call to saved contact	PASS
Network Loss Fallback	Wi-Fi and mobile data turned off on phone	On-Device Inertial Fallback triggers locally; siren and SMS/Call fire	PASS
High Concurrency	Mobile app streams 6 batches/sec	`ThreadingHTTPServer processes non-blocking; 0 timeouts; 200 buffer`	PASS